VCS Methodology

VM0042

IMPROVED AGRICULTURAL LAND MANAGEMENT

Version 2.1

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Sectoral Scope 14

Version 1.0 of this methodology was developed by TerraCarbon LLC and Indigo Ag. The lead authors were David Shoch and Erin Swails from TerraCarbon. Contributions from Indigo were made by (in alphabetical order): Chris Black, Charlie Brummit, Nell Campbell, Max DuBuisson, Dan Harburg, Lauren Matosziuk, Melissa Motew, Guy Pinjuv, and Ed Smith. Version 1.0 was approved on 19 October 2020.

Valuable input was provided by: Peter Beare (World Business Council for Sustainable Development, Switzerland), Deborah Bossio (The Nature Conservancy, USA), Rori Cowan (Climate Smart Group, USA), Annette Cowie (New South Wales Department of Primary Industries, Australia), Jessica Davies (Lancaster University, UK), Karen Haugen-Kozyra (Viresco Solutions, Canada), Louisa Kiely (Carbon Farmers of Australia, Australia), Johannes Lehmann (Cornell University, USA), Paul Luu (4 per 1000 Initiative, France), Ken Newcombe (C-Quest Capital, USA), Sean Penrith (Gordian Knot Strategies, USA), Jeffrey Seale (Bayer US – Crop Science, USA), Tom Stoddard (Native Energy, USA), Moritz von Unger (Silvestrum Climate Associates, LLC, USA), Matthew Warnken (Agriprove: Soil Carbon Solutions, Australia), and Leigh Winowiecki (World Agroforestry, Kenya).

Version 2.0 of this methodology was prepared by Verra. Revisions to the uncertainty section were prepared by Dan Kane and Jaclyn Kachelmeyer, TerraCarbon LLC with input from Brian McConkey, Viresco Solutions and Beth Ziniti, Applied Geosolutions; and in consultation with several external experts.

Valuable input was provided by: Denis Angers (Laval University, Canada), Eyal Ben-Dor (Tel Aviv University, Israel), Charlie Brummit (Indigo Ag, USA), Rich Conant (Colorado State University, USA), Ciniro Costa Jr. (Alliance of Bioversity International and the International Center for Tropical Agriculture, Brazil), Annette Cowie (New South Wales Department of Primary Industries, Australia), Cole D. Gross (University of Alberta, Canada), Mario Guevara (National Autonomous University of Mexico, Mexico), Matthew Harrison (University of Tasmania), Stefan Hauser (International Institute of Tropical Agriculture, Nigeria), Beverly Henry (Queensland University of Technology, Australia), Dan Kane (TerraCarbon, USA), Tony Knowles (Cirrus Group, South Africa), Emily Kyker-Snowman (Yale University, USA), Johannes Lehmann (Cornell University, USA), Jose Lucas Safanelli (Woodwell Climate Research Center, USA), Brian McConkey (Viresco Solutions, Canada), Emily Oldfield (EDF, USA), Guillermo Peralta (FAO/Carbon Group, Argentina), Cornelia Rumpel (INRAE, France), Jonathan Sandermann (Woodwell Climate Research Center, USA), Pete Smith (University of Aberdeen, Scotland), Adam von Haden (University of Wisconsin, USA), Britta Weber (Ruumi, Germany), John Wendt (International Fertilizer Development Center, Kenya), Niklas Witt (Klim, Germany), and Beth Ziniti (Regrow, USA).

CONTENTS

1 SOURCES 5

2 SUMMARY DESCRIPTION OF THE METHODOLOGY 5

3 DEFINITIONS 7

4 APPLICABILITY CONDITIONS 9

5 PROJECT BOUNDARY 11

6 BASELINE SCENARIO 13

7 ADDITIONALITY 17

1 SOURCES

This methodology is based on the following methodologies:

- *VM0017 Adoption of Sustainable Agricultural Land Management, v1.0*
- *VM0022 Quantifying N_2O Emissions Reductions in Agricultural Crops through Nitrogen Fertilizer Rate Reduction, v1.1*
- *VM0026 Methodology for Sustainable Grassland Management, v1.1*

This methodology uses the latest versions of the following Clean Development Mechanism (CDM)methodologies and tools:

- *AR-TOOL14 Methodological Tool: Estimation of Carbon Stocks and Change in Carbon Stocks of Trees and Shrubs in A/R CDM Project Activities*
- *Tool for Testing Significance of GHG Emissions in A/R CDM Project Activities*[1]
- *TOOL24 Methodological Tool: Common Practice*
- *A/R Methodological Tool: Tool for the Identification of Degraded or Degrading Lands for Consideration in Implementing CDM A/R Project Activities*[2]
- *TOOL16 Methodological Tool: Project and Leakage Emissions from Biomass*

2 SUMMARY DESCRIPTION OF THE METHODOLOGY

Table 1: Additionality and crediting baseline methods

Additionality and Crediting Method	
Additionality	Project Method
Crediting Baseline	Project Method

This agricultural land management (ALM) methodology provides procedures to estimate the greenhouse gas (GHG) emission reductions (CO_2, CH_4, and N_2O) and carbon dioxide removals

[1] This tool was deactivated by the CDM as it is no longer required in light of methodology improvements (see CDM Executive Board Meeting Report 68 from 16–20 July 2012). There were no technical concerns with the procedures described in the tool; therefore it is still valid in the context of VM0042.

[2] This tool was deactivated by the CDM as it is no longer required as a standalone document (see CDM Executive Board Meeting Report 75 from 30 September–4 October 2013). There were no technical concerns with the procedures described in the tool; therefore it is still valid in the context of VM0042.

(reductions and removals) resulting from the adoption of improved ALM practices. The methodology is compatible with regenerative agriculture and has a particular focus on increasing soil organic carbon (SOC) storage.

The crediting baseline and additionality are determined via a project method (Table 1). The baseline scenario assumes the continuation of pre-project ALM practices. Practices in the baseline scenario are determined by applying a minimum three-year historical look-back period to produce an annual schedule of activities (i.e., tillage, planting, harvest, and fertilization events) for each quantification unit within the project area (e.g., for each field), to be repeated over the baseline period.[3] Baseline emissions/stock changes are then modeled. Alternatively, baseline SOC stock change may be directly measured in "baseline control sites" managed according to pre-project practices as set out in the schedule of activities. The baseline scenario is re-evaluated as required by the most recent version of the *VCS Standard*, and revised, where necessary, to reflect current agricultural production in the region.

Additionality is demonstrated by a barrier analysis and showing that the practice change implemented under the project activity is not common practice. A practice change constitutes any of the following:

- Adoption of a new practice (e.g., adoption of one or more of the practices covered in the categories included in Applicability Condition 1);
- Cessation of a pre-existing practice (e.g., stop tillage or irrigation);
- Adjustment to a pre-existing practice; or
- Some combination of the above.

Any quantitative adjustment (e.g., decrease in fertilizer application rate) must exceed 5% of the pre-existing value to qualify as a practice change.

The methodology provides three approaches to quantifying reductions and removals resulting from the adoption of improved ALM practices.

Quantification Approach 1: Measure and Model – a biogeochemical, process-based model is used to estimate GHG fluxes related to SOC stock changes, soil methanogenesis, and use of nitrogen fertilizers and nitrogen-fixing species. Edaphic characteristics and actual agricultural practices implemented, measured initial SOC stocks, and climatic conditions in sample fields are used as model inputs. Periodic measurements of SOC stocks are required every five years at minimum (see Table 8).

Quantification Approach 2: Measure and Re-Measure – direct measurement is used to quantify changes in SOC stocks. This approach is relevant where models are unavailable or have not yet been validated or parameterized for a particular region, crop, or practice, or where project proponents prefer to use a direct measurement approach for SOC stock change. Quantification

[3] Project proponents must periodically reassess ALM project baselines. See the most recent version of the *VCS Standard* for further details on baseline re-assessment requirements.

Approach 2 directly measures SOC stock changes in the baseline scenario in linked baseline control sites.

Quantification Approach 3: Default Factors – CO_2 flux from fossil fuel combustion and N_2O and CH_4 fluxes, excluding CH_4 flux from methanogenesis, are calculated using default emission factors.

The quantification approach varies by emission/removal type. Approaches to quantification of contributing sources of CO_2, CH_4, and N_2O emissions are listed in Table 5.

3 DEFINITIONS

In addition to the definitions set out in the VCS document *Program Definitions*, the following definitions apply to this methodology.

Annual
A plant species that within one year completes its life cycle, reproduces, and dies.

Baseline control site
Defined area that is managed according to pre-project (baseline) practices (as set out in the schedule of activities) for direct measurement of baseline soil organic carbon stock change. It is linked to and representative of the land in one or more quantification units. Baseline control sites may be within or outside of the project area.

Historical look-back period
The time period prior to the project start date covering at minimum three years and one complete crop rotation. The historical look-back period is used to produce the schedule of activities (see definition below).

Improved agricultural land management practice
An agricultural practice yielding increased soil organic carbon storage or other climate benefit, involving a refinement to fertilizer or other soil amendment application, water management/irrigation, tillage, residue management, crop planting and harvesting, and/or grazing practices. Projects may implement a single practice or a combined (stacked) practice on each field or land part of their project.

Inelastic neutron scattering (INS)
An in-field (in situ) measurement technique based on the detection and analysis of gamma rays emitted by soil elements after irradiation with neutrons. It is also known as neutron-stimulated gamma ray analysis or spectroscopy.

Infrared spectroscopy
Mid-infrared (MIR), near-infrared (NIR), and visible near-infrared (Vis-NIR) spectroscopy, including diffuse reflectance spectroscopy (DRS) and diffuse reflectance infrared Fourier

transform spectroscopy (DRIFT). Vis-NIR combines the visible and near-infrared electromagnetic range and usually refers to a wavelength range from 350 to 2500 nm (visible range is between 350 and 700 nm). MIR covers the range between 4000 cm^{-1} and 600 (or 400) cm^{-1}, depending on the instrument.

Laser-induced breakdown spectroscopy (LIBS)
Application of a high-energy pulse to soil samples to generate a high-temperature plasma, which emits radiation at different wavelengths depending on the elements present in the sample

Nitrogen-fixing species
Any plant species that associates with nitrogen-fixing microbes found within nodules formed on the roots, including but not limited to soybeans, alfalfa, and peas

Organic nitrogen fertilizer
Any organic material containing nitrogen, including but not limited to animal manure, compost, and biosolids

Perennial
A plant species whose life cycle, reproduction, and death extends across multiple years

Professional agronomist
An individual with specialized knowledge, skills, education, experience, or training in crop and/or soil science. Such individuals may be agricultural experts like soil scientists, husbandry specialists, agronomists, or representatives of a governmental agricultural body.

Project domain
Set of conditions (including crop type, soil texture, and climate) in which model application has been validated (see *VCS Module VMD0053 Model Calibration, Validation, and Uncertainty Guidance for Improved Agricultural Land Management*)

Sample point
Sample location of undefined area

Quantification unit
Defined area within the project for which GHG emission reductions and carbon dioxide removals (reductions and removals) are estimated using the selected quantification approach.

Schedule of activities
Annual schedule of historical management/activity practices applied in the baseline scenario over the historical look-back period (e.g., tillage, planting, harvest, and fertilization events). These practices are determined following the data requirements given in Box 1.

Stratum
A subset of each quantification unit within which the value of a variable, and the processes leading to change in that variable, are relatively homogenous.

Synthetic nitrogen fertilizer

Any fertilizer made by chemical synthesis (solid, liquid, gaseous) and containing nitrogen. This may be a single nutrient fertilizer product (only including N), or any other synthetic fertilizer containing N, such as multi-nutrient fertilizers (e.g., N–P–K fertilizers) and "enhanced-efficiency" N fertilizers (e.g., slow release, controlled release, and stabilized N fertilizers).

Woody perennials

Trees and shrubs having a life cycle lasting more than two years, excluding cultivated annual species with lignified tissues, such as cotton or hemp

4 APPLICABILITY CONDITIONS

This methodology applies to a broad range of project activities that increase SOC storage and/or decrease net emissions of CO_2, CH_4, and N_2O from ALM operations compared to the baseline scenario. The methodology is globally applicable.

This methodology is applicable under the following conditions:

1) Projects introduce or implement one or more new changes to pre-existing ALM practices which:
    a) Improve fertilizer (organic or inorganic) management;
    b) Improve water management/irrigation;
    c) Reduce tillage/improve residue management;
    d) Improve crop planting and harvesting (e.g., improved agroforestry, crop rotations, cover crops); and/or
    e) Improve grazing practices.

    Appendix 1 provides a non-exhaustive list of eligible ALM practices. A change in practice constitutes adoption of a new practice, cessation of a pre-existing practice, or adjustment to a pre-existing practice that results in reductions or removals.

2) Projects introduce or implement quantitative adjustments (e.g., decrease in fertilizer application rate) that exceed 5% of the pre-existing value, calculated as the average value over the historical look-back period, developed for the baseline schedule of activities (see Section 6). Appendix 1 gives additional details and guidance on practices and on determining practice change.
3) Project activities are implemented on land that is either cropland or grassland at the project start date. The land must remain cropland or grassland throughout the project lifetime except under the following scenarios:
    a) Introduction of temporary grassland into cropland where it is demonstrated, prior to the project start date and to the addition of new project activity instances, that the integration of forage crops (e.g., annual/perennial grasses, legumes) into

annual crops is part of a planned, long-term ALM system (e.g., integrated crop-livestock system). Project proponents must provide documentation of the long-term management plans, covering the duration of the project, that describe proposed practices, crops, and expected benefits and outcomes of integrated grassland-cropland management; or

b) A one-time conversion from grassland to cropland or vice versa where it is demonstrated, prior to project validation, that project lands in the baseline scenario are degraded and the introduction of improved land use change practices would significantly improve soil health. Project proponents must provide documentation demonstrating that lands are degraded at the start of the project and degradation will continue in the baseline scenario due to the presence of degradation drivers or pressures in the baseline scenario. See Appendix 2 for procedures on how to propose this type of land use change.

4) Empirical or process-based models used to estimate stock change/emissions via Quantification Approach 1 are:

a) Publicly available, though not necessarily free of charge,[4] from a reputable and recognized source (e.g., the model developer's website, IPCC, or government agency). Sufficient conceptual documentation of inputs, outputs, and information on how the model functionally represents SOC dynamics must be accessible to the public. The project proponent must include the model source in the project description (e.g., hyperlink to the model and date of webpage access or citation of peer-reviewed publication). Providing the source code or an API for independent replication of calculations is not required;

b) Shown in peer-reviewed scientific studies to successfully simulate changes in SOC and trace gas emissions resulting from the changes in ALM practices included in the project description;

c) Able to support repetition of the project model simulations. This includes clear versioning of the model used in the project and stable software support, as well as fully reported sources and values for all parameters used with the project version of the model. Where multiple sets of parameter values are used in the project, clearly identify the sources of varying parameter sets and how they were applied to estimate stock change/emissions in the project. Acceptable sources include peer-reviewed literature and statements from appropriate expert groups that demonstrate evidence of expertise with the model via authorship of peer-reviewed model publications or authorship of reports for entities supporting climate-smart agriculture. These sources must describe the datasets and statistical processes used to set parameter values;

d) Validated per datasets and procedures detailed in Section 5.2 of *VCS Module VMD0053 Model Calibration, Validation, and Uncertainty Guidance for Improved Agricultural Land Management.* Model prediction error must be calculated using

[4] It is the responsibility of the project proponent to ensure they have any required licenses for models used in a project.

datasets as described in Section 5.2.5 of *VMD0053* and must use the same parameters or sets of parameters applied to estimate stock change/emissions in the project; and

e) Using the same model version in the baseline and project scenarios. Further, the same parameters/parameter sets must be used in the baseline and project scenarios. Model input data must be derived following guidance in Table 6 and Table 8. Model uncertainty must be quantified following guidance in Section 8.6.

This methodology is not applicable under the following conditions:

5) The project area has been cleared of native ecosystems within the 10-year period immediately prior to the project start date.
6) The project activity is expected to cause a sustained reduction in productivity of greater than 5%, as demonstrated by peer-reviewed and/or published studies on activity in the region or a comparable region.
7) The project activity is biochar application. Biochar may be applied as a soil amendment in the project area provided that the total organic carbon content[5] of the biochar applied is subtracted from the estimated SOC stock change in the project scenario at each verification event.
8) The project activities occur on a wetland; this condition does not exclude crops subject to artificial flooding where it is demonstrated that crop cultivation does not impact the hydrology of any nearby wetlands.

5 PROJECT BOUNDARY

The spatial extent of the project boundary is all lands planning to implement the proposed improved ALM practice(s). Carbon pools included in the project boundary in the baseline and project scenarios are listed in Table 2.

Table 2: Selected carbon pools in the baseline and project scenarios

Source	Included?	Justification/Explanation
Aboveground woody biomass	Yes / Optional	Aboveground woody biomass must be included where project activities significantly reduce the pool compared to the baseline. In all other cases, aboveground woody biomass is an optional pool.
Aboveground non-woody biomass	No	Carbon pool is not included because it is not subject to significant changes or potential changes are transient in nature.

[5] To estimate the total carbon content of the applied biochar, project proponents should follow the procedures set out in Sections 8.2.2.1 or 8.2.2.2 (for high- or low-technology production facilities, respectively) in the most recent version of *VM0044 Methodology for Biochar Utilization in Soil and Non-Soil Applications*. Where the technology production facility type is not known, procedures in the low-technology approach (Section 8.2.2.2) should be followed for conservativeness.

Belowground woody biomass	Optional	Belowground woody biomass may optionally be included where project activities significantly increase the pool compared to the baseline.
Belowground non-woody biomass	No	Carbon pool is not included because it is not subject to significant changes or potential changes are transient in nature.
Dead wood	No	Carbon pool is not included because it is not subject to significant changes or potential changes are transient in nature.
Litter	No	Carbon pool is not included, because it is not subject to significant changes or potential changes are transient in nature.
SOC	Yes	Major carbon pool affected by project activity that is expected to increase in the project scenario
Wood products	No	Carbon pool is optional for ALM project methodologies and may be excluded from the project boundary.

GHG sources included in the project boundary in the baseline and project scenarios are listed in Table 3. Specific carbon pools and GHG sources may be deemed de minimis and need not be accounted for (i.e., value set to zero) where together the omitted decrease in carbon stocks (in carbon pools) or increase in GHG emissions (from GHG sources) amounts to less than 5% of the total GHG benefit generated by the project. This includes sources and pools that cause project and leakage emissions. This and all subsequent references to de minimis demonstration are conducted via application of the CDM *Tool for testing significance of GHG emissions in A/R CDM project activities.*[6] The SOC pool must be included in the project boundary[7] (i.e., it must be monitored as part of a *VM0042* project and is not allowed to be deemed de minimis).

Table 3: GHG sources included in or excluded from the project boundary in the baseline and project scenarios

Source	Gas	Included?	Justification/Explanation
SOC	CO_2	Yes	Quantified as stock change in the pool, rather than an emissions source (see Table 2)
Fossil fuel	CO_2	S*	Sources of fossil fuel emissions are vehicles (mobile sources, such as trucks, tractors) and mechanical equipment required by the ALM activity.
Liming	CO_2	S*	Application of limestone or dolomite as soil amelioration may represent a significant source of CO_2.
Soil methanogenesis	CH_4	S*	Anoxic conditions in soils may lead to soil methanogenesis.

[6] Since project activities must not result in a sustained reduction in productivity (including animal weight gains) or sustained displacement of any pre-existing productive activity, feedlots are conservatively excluded from the project boundary.

[7] Note that the methodology does not separately account for SOC stock changes resulting from erosion.

Enteric fermentation	CH_4	Yes	Where livestock are present in the project or baseline scenarios, CH_4 emissions from enteric fermentation must be included in the project boundary.
Manure deposition	CH_4	Yes	Where livestock are present in the project or baseline scenarios, CH_4 and N_2O emissions from manure deposition and management must be included in the project boundary.
	N_2O	Yes	
Use of nitrogen fertilizers	N_2O	Yes	Must be included where: • The project area would have been subject to nitrogen fertilization in the baseline scenario; or • Nitrogen fertilization is greater in the with-project scenario relative to the baseline scenario.
Use of nitrogen-fixing species	N_2O	Yes	Where nitrogen-fixing species are planted in the project, N_2O emissions from nitrogen-fixing species must be included in the project boundary.
Biomass burning	CO_2	Excluded	Carbon stock decreases due to burning are accounted as a carbon stock change.
Biomass burning	CH_4	S*	Biomass burning releases CH_4.
	N_2O	S*	Biomass burning releases N_2O.
Woody biomass	CO_2	S*	Quantified as stock change in the pool rather than an emissions source (see Table 2)

S* – Must be included where the project activity significantly increases emissions (i.e., by more than 5%) compared to the baseline scenario and may be included where the project activity reduces emissions compared to the baseline scenario.

6 BASELINE SCENARIO

Continuation of pre-project ALM practices is the most plausible baseline scenario. For each quantification unit (e.g., for each field), baseline scenario practices are set to match the practices implemented in the historical look-back period, creating an annual schedule of activities to be repeated throughout the first baseline period.[8] Baseline emissions/stock changes are then modeled (Quantification Approach 1) or (for SOC stock change only) directly measured in baseline control sites subject to the annual schedule of activities (Quantification Approach 2). Note that under Quantification Approach 1, direct SOC stock estimates are also required at time t = 0 years to serve as model input for model initialization.[9] The crops and practices assumed in the baseline scenario must be re-assessed in accordance with the

[8] For example, where the schedule of activities includes tillage events in years $t = -3$ and -1 but does not involve tillage in year $t = -2$, the schedule of activities for tillage in the baseline scenario would be tillage, no tillage, tillage. This pattern would be repeated as follows for the first baseline period: tillage, no tillage, tillage, tillage, no tillage, tillage, tillage, no tillage, tillage, tillage.

[9] Per Table 6, baseline SOC stocks may be (back-)modeled to t = 0 from measurements collected within ±5 years of t = 0.

requirements of the most recent version of the *VCS Standard* and revised, where necessary, to reflect current agricultural production in the region.[10]

Development of Schedule of Activities in the Baseline Scenario

For each quantification unit, a schedule of activities in the baseline scenario is determined by assessment of practices implemented during the period prior to the project start date. The interval over which practices are assessed, x years, must be a minimum of three years and must include at least one complete crop rotation, where applicable. Where a crop rotation is not implemented in the baseline, $x \geq 3$ years. For each year, $t = -1$ to $t = -x$ (i.e., years preceding project start), information on ALM practices must be determined, per the requirements presented in Table 4.

In most cases, quantitative information is associated with related qualitative information (see Box 1). Thus, a negative response on a qualitative element would mean there is no quantitative information related to that practice, whereas a positive response on a qualitative element requires quantitative information related to that practice.

The schedule of activities, beginning with year $t = -x$, is applied in the baseline scenario, from $t = 1$ onward, repeating every x years through the end of the first baseline period.

The schedule of activities in the baseline scenario is valid until re-assessment is required as per the most recent version of the *VCS Standard*. At the end of each baseline period, production of the commercial crop(s) in the baseline scenario is re-evaluated. Published regional (sub-national) agricultural production data from within the five years immediately preceding the end of the current baseline period must be consulted.

- Where there is evidence of continued production of the relevant commercial crop(s) using the same ALM practices in the region, the baseline scenario is valid as-is, continuing with the previous schedule of activities.

- Where there is no evidence of continued production of the relevant commercial crop(s), a new schedule of ALM activities (evaluated against common practices in the region) is developed based on written recommendations for the sample field provided by independent professional agronomists, agricultural experts such as soil scientists, husbandry specialists, and agronomists, or representatives of a governmental agricultural body, including government agricultural extension agents. Recommendations must provide sufficient detail to produce the minimum specifications on ALM practices for the baseline scenario as outlined in Table 4.

- Where more than one value is documented in recommendations (e.g., where a range of application rates are prescribed in written recommendations), the principle of conservativeness must be applied, selecting the value that results in the lowest expected emissions (or highest rate of stock change) in the baseline scenario.

[10] See Section 3.2.7 of the *VCS Standard, v4.7*.

- Where the evidence is not field-specific, conservatively derived field-specific values must be supported by a documented method justifying the appropriateness of selection.

Table 4: Minimum specifications for ALM practices in the baseline scenario

ALM Practice	Qualitative	Quantitative
Crop Planting and Harvesting	• Crop type(s)	• Approximate date(s) planted (where applicable) • Approximate date(s) harvested/terminated (where applicable) • Crop yield (where applicable)
Nitrogen Fertilizer Application	• Manure (Y/N) • Compost (Y/N) • Synthetic N fertilizer (Y/N)	• Manure type application rate (where applicable) • Compost type application rate (where applicable) • N application rate in synthetic fertilizer (where applicable)
Tillage and/or Residue Management	• Tillage (Y/N) • Crop residue removal (Y/N)	• Depth of tillage (where applicable) • Frequency of tillage (where applicable) • Percent of soil area disturbed (where applicable) • Percent of crop residue removed (where applicable)
Water Management/Irrigation	• Irrigation (Y/N) • Flooding (Y/N)	• Irrigation rate (where applicable)
Grazing Practices	• Grazing (Y/N) • Animal type (where applicable) • Harvesting/mowing (Y/N)	• Animal stocking rate (i.e., number of animals and length of time grazing in each area annually, where applicable) • Frequency of harvest
Liming	• Application of calcitic limestone or dolomite (Y/N)	• Calcitic limestone or dolomite application rate (where applicable)

*Y/N: Yes/No

Box 1: Sources of qualitative and quantitative data

Sources of information for all undefined activity/management-related model input variables (see Table 6 and Table 8), and all parameters relevant to the baseline scenario (i.e., all parameters with the subscript *bsl* that reference Box 1 in its respective parameter table), must follow the requirements detailed below.

All qualitative information on ALM practices must be determined via consultation with, and substantiated with a signed attestation from, the farmer or landowner of the sample field during that period. Where the farmer or landowner is not able to provide qualitative information (e.g., a sample field is newly leased), the project proponent must follow the quantitative information hierarchy outlined below.

The following list specifies the allowable sources of quantitative information on ALM practices in descending order of preference, as available:

1) Historical management records supported by one or more forms of documented evidence pertaining to the selected sample field and period $t = -1$ to $t = -3$ (e.g., management logs, receipts or invoices, farm equipment specifications, logs or files containing machine and/or sensor data) or remote sensing (e.g., satellite imagery, manned aerial vehicle footage, drone imagery), where requisite information on ALM practices can be reliably determined with these methods (e.g., tillage status, crop type, irrigation)

2) Historical management plans supported by one or more forms of documented evidence pertaining to the selected sample field and period $t = -1$ to $t = -3$ (e.g., management plan, recommendations in writing solicited by the farmer or landowner from an agronomist). Where more than one value is documented in historical management plans (e.g., where a range of application rates are prescribed in written recommendations), the principle of conservativeness must be applied and the value that results in the lowest expected emissions (or highest rate of stock change) in the baseline scenario must be selected.

3) A signed attestation from the farmer or landowner of the sample field during that period – where the attested value does not deviate significantly from other evidence-supported values for similar fields (e.g., fertilizer data from adjacent fields with the same crop, adjacent years of the same field, government data on application rates in that area, or statement from a local extension agent regarding local application rates). The VVB must determine whether the data are sufficient. In circumstances where this requirement is not met, Option 4 must be followed.

4) Regional (sub-national) average values derived from agricultural census data or other sources from within the 20-year period preceding the project start date or the 10 most recent iterations of the dataset, whichever is more recent. Where estimates have been disaggregated by relevant crop or ownership classes, those should be used. The estimates must be substantiated with a signed attestation from the farmer or landowner of the sample field during that period. Examples include the USDA National Agricultural Statistics Service Quick Stats database and USDA Agricultural Resource Management Survey.

This hierarchy applies to any additional quantitative inputs required by the model (Quantification Approaches 1 and 2) or default factor (Quantification Approach 3) selected. The principle of conservativeness must be applied in all cases.

7 ADDITIONALITY

This methodology uses a project method for the demonstration of additionality. For grouped projects, demonstration of additionality must be based upon the initial project activity instances (see further details in the latest version of the *VCS Standard*). Project proponents using this methodology must:

1) Demonstrate regulatory surplus;

2) Identify institutional barriers that would prevent the implementation of a change in pre-existing ALM practices; and

3) Demonstrate that adoption of the suite of proposed project activities is not common practice.

Further details on each of these steps are provided below.

Step 1: Regulatory surplus

The project proponent must demonstrate regulatory surplus in accordance with the rules and requirements set out in the most recent version of the *VCS Standard*.

Step 2: Identify institutional barriers that would prevent implementation of a change in pre-existing ALM practices

The project proponent must determine whether there are cultural and/or social barriers (e.g., cultural practices and social norms, attitudes, and beliefs) to the proposed change(s) in ALM expected that prevent implementation of the change without the project proponent's intervention and the resulting revenue from the sale of Verified Carbon Units (VCUs). The project proponent must list and describe barriers to the implementation of proposed changes to pre-project ALM practices to establish that the change would not occur if the project was not undertaken by the project proponent and registered as a VCS project.

Demonstration of cultural and/or social barriers must be supported by peer-reviewed and/or published studies specific to the project region. Where evidence is not available for the project region, evidence from other regions may be used where it is demonstrated how those cultural and/or social barriers also apply to the project region.

Such barriers may include traditional knowledge or lack thereof, laws and customs, market conditions, and lack of motivating incentives to change practices, including, but not limited to:

- Traditional equipment and technology
- Grower risk tolerance and beliefs about the feasibility of adopting new practices, and implications of their decisions

- Grower openness to new ideas and perceptions of the magnitude of the change
- Barriers associated with grower identity

Step 3: Demonstrate that adoption of the suite of proposed project activities is not common practice

The project proponent must determine whether the proposed project activity or suite of activities[11] are common practice in each region (or 'geographic area' as defined for grouped projects in the *VCS Standard*) included within the project area. Common practice is defined as greater than 20% adoption.[12] Where the project is implementing only one activity, the adoption rate of that practice must be lower than 20 percent in each region within the project spatial boundary.

To demonstrate that a project activity or suite of activities is not common practice, the project proponent must show that the area-dependent weighted mean adoption rate of the two (or more) predominant[13] proposed project activities within each region covered by the project area is below 20%[14] (see Equation (1)). Therefore, in projects where the adoption rate of one activity (e.g., reduced tillage) is greater than 20%, the project must include a proportionally higher ratio of other activities with lower adoption rates (e.g., cover crops, improved fertilizer management) to bring the weighted average of proposed project activities below 20%. An individual activity with an existing adoption rate in the relevant region below or equal to 20% is always considered additional. An individual activity with an existing adoption rate greater than 20% may only be considered additional through the assessment of the weighted mean adoption rate for all lands part of the project area within that region.

Categories of project activities for the demonstration of common practice may be defined according to the categories in the evidence provided, or using the categories outlined in Appendix 1.

Evidence must be provided in the form of publicly available information contained in:

a) Agricultural census or other government (e.g., survey) data;

b) Peer-reviewed scientific literature;

c) Independent research data; or

[11] The suite of activities refers to all activities implemented across the aggregated project. It does not refer to the activities implemented on each individual farm.

[12] Twenty percent is the precedent for a common practice threshold established in Section 18 of the CDM *Methodological tool: Common practice*. Available at: https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf

[13] Determined based on the extent of the project area (i.e., hectares) covered.

[14] Where a project is planning to implement two activities, common practice must be assessed based on the weighted mean of those two activities. Where only one activity is implemented, common practice must be assessed solely based on that activity's adoption rate (i.e., the adoption rate of that activity must be below 20%).

d) Reports or assessments compiled by industry associations.

To demonstrate common practice, the project area must be stratified to the state or provincial level (or equivalent second-order jurisdiction) in the countries where the project is being developed. Where supporting evidence is unavailable at the state/provincial level (e.g., in developing countries), aggregated data or evidence at a subnational, national, or regional level may be used with justification. Where stratification based on geopolitical boundaries is impractical (e.g., due to lack of data), other forms of stratification, such as major soil types or cropping zones, may be used with justification. The same stratification approach and data sources must be applied across the entire project to maintain the integrity of the common practice demonstration. Where a common practice assessment data source is unavailable for a subset of the project region, justification must be provided to use a different data source.

Where evidence for a single proposed project activity in the region is not available from any of these sources, the project proponent may obtain a signed and dated attestation statement from a qualified independent local expert (e.g., agricultural extension agent, accredited agronomist) estimating the adoption rate for the weighted mean calculation. Where evidence on the suite of proposed activities is unavailable, a qualified independent local expert may provide a signed and dated attestation statement stating whether the proposed suite of project activities is common practice in the region and providing estimated values for the regional existing adoption rate of proposed project activities (EA_{ay}; see Equation (1)). All projects using independent local expert attestation must provide the qualifications of the expert and the methods used to inform their analysis.

Equation (1) must be applied to calculate the weighted mean adoption rate in each region included in the project area. In case the project involves only one activity, and the project proponent demonstrates that the adoption rate of that activity is below 20% in the region where the project area is located, Equation (1) is not required.

$$AR = (EA_{a1} \times PA_{a1}) + (EA_{a2} \times PA_{a2}) + \cdots + (EA_{ay} \times PA_{ay}) \quad (1)$$

Where:[15]

$$PA_{a1} = \frac{Area_{a1}}{(Area_{a1} + Area_{a2} + \cdots + Area_{ay})}$$

$$PA_{a2} = \frac{Area_{a2}}{(Area_{a1} + Area_{a2} + \cdots + Area_{ay})}$$

$$PA_{ay} = \frac{Area_{ay}}{(Area_{a1} + Area_{a2} + \cdots + Area_{ay})}$$

and:

[15] Note that parameters are described below equations only at their first appearance.

AR	=	Weighted mean adoption rate in each region included in the project area (%)
EA_{ay}	=	Existing adoption rate of proposed project activity *ay* in the region (%)
PA_{ay}	=	Ratio of proposed project-level adoption of activity *ay* relative to sum of proposed project-level adoption of all activities in the project area
$Area_{ay}$	=	Area of proposed project-level adoption of activity *ay* in the project area (hectares)
ay	=	1, …, *ay* proposed project activities

A project proponent may include areas where more than one project activity will be implemented on the same land (e.g., reduced tillage plus cover crops). Evidence of existing adoption rates for the combined (two or more) activities should be used to calculate the weighted mean adoption rate of the proposed combined activities (i.e., parameter *EAay* would represent the adoption rate of a combined activity as the two are implemented on the same land or field). Where evidence on existing adoption rates for the combined activities is not available, the project proponent may multiply the existing adoption rates (i.e., pre-project) of the individual activities to estimate the combined activity adoption rate.[16] For example, with a statewide existing adoption rate of 40% for reduced tillage and 10% for cover-cropping, the adoption rate to be applied in Equation (1) for lands combining (stacking) these two activities would be 4% (i.e., *EAay* = 0.4 × 0.1 = 0.04).

Where Steps 1–3 are satisfied, the proposed project activity is additional.

For registered grouped projects with an initial set of project activity instances, Appendix 3 provides a recommended process for assessing whether new project activity instances are common practice.

[16] In practice, this encourages "stacking" of new activities to enhance reductions and/or removals compared to implementing only one new activity on a given area or farm.